

Syed Musadiq Hussain

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SUMMARY

Early-career AI/ML Developer with hands-on project experience designing and building Retrieval-Augmented Generation (RAG) pipelines, Large Language Model (LLM) integrations, and end-to-end automation systems. Skilled in Python development, API design, vector search, containerization, and cloud deployment (GCP). Strong foundation in prompt engineering, NLP concepts, and applied machine learning, with a continuous focus on MLOps practices, scalable system design, and production-ready AI solutions.

TECHNICAL SKILLS

Languages & Core	Python, SQL, Bash/Linux, Git, OOP, REST API Design
AI/ML & GenAI	LLMs, RAG, NLP, Prompt Engineering, LangChain, Hugging Face, OpenAI API, Vector Search, basic Fine-tuning concepts
Frameworks	PyTorch (learning), Scikit-learn, Pandas, NumPy, FastAPI, Flask, Streamlit
Data & Storage	PostgreSQL, MySQL, MongoDB, Vector Databases (Chroma, FAISS)
Cloud & DevOps	Docker, Git/GitHub, GCP (Cloud Run, Cloud Secret Manager, Vertex AI basics), CI/CD (GitHub Actions)
Automation & Integration	n8n, Make.com, REST APIs, Webhooks, Asynchronous Programming

PROJECTS

RAG-Chat – Context-Aware AI Assistant

- Built a Retrieval-Augmented Generation (RAG) document Q&A pipeline using LangChain, Chroma, and FAISS for semantic search and retrieval over private documents.
- Added source citation and grounding in responses to reduce hallucinated answers and improve output reliability and trust.
- Containerized the application with Docker, deployed it on GCP Cloud Run, and set up automated CI/CD with GitHub Actions.

AI-Voice-WhatsApp-Helper

- Built a conversational AI voice assistant for WhatsApp using Twilio, OpenAI Whisper for speech-to-text transcription, and Google Gemini 2.0 for natural language responses, with Serper for real-time web search augmentation.
- Implemented an asynchronous FastAPI backend with rate limiting, structured error handling, and request logging for reliability.
- Deployed on Google Cloud Platform with secrets securely managed via GCP Secret Manager.

EXPERIENCE

Freelance AI/ML Developer

2025 – Present

- Built small AI tools (chatbots, simple automation workflows) for individual clients using Python, LangChain, and FastAPI.
- Set up basic data pipelines to pull and clean data from REST APIs and SQL sources for use in client projects.
- Used Docker and GCP to deploy and host project demos for clients.

EDUCATION

High School Diploma – Islamia College Peshawar
Self-study / Online Courses – Virtual University of Pakistan

2024
Ongoing

CERTIFICATIONS

Google Cloud Certified – Professional Cloud Architect (cloud infrastructure design fundamentals)

2025